# Deepthi Karani

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## Education

**Carnegie Mellon University — School of Computer Science** **December 2027**
*Master of Computational Data Science* **GPA: 4.0/4.0**
- **Coursework**: ML in Production, Advanced NLP, Machine Learning, Data Science Seminar (F26). Pittsburgh, PA

**PES University** **May 2026**
*Bachelor of Technology in Computer Science and Engineering* **GPA: 9.12/10.00**
- **Coursework**: Data Structures, OOP, Operating Systems, Cloud Computing, Big Data, GenAI. Bangalore, India
- **Teaching Assistant** for i) Python for Computational Problem Solving, ii) Database Management Systems.

## Skills

**Languages:** Python, C++, Java, SQL, JavaScript, HTML/CSS
**ML / AI**: PyTorch, TensorFlow, scikit-learn, XGBoost, LangGraph, RAG, LLM Agents, MCP, NLP, Explainable AI
**Data & Analytics**: Pandas, NumPy, SQL, Apache Kafka, EDA & Feature Engineering, Predictive Modeling
**Cloud & Systems**: Azure ML, Docker, Kubernetes, Linux, Git, REST/OpenAPI, Google Colab

## Experience

***SAP***
**Software Developer Intern** **January - July 2026**
- Engineered a low-latency **MCP gateway** for SAP Integration Suite, exposing backend **enterprise APIs** as discoverable, invocable **tools/resources for LLM agents**, translating REST/OpenAPI service contracts into a standardized protocol layer.
- Built API-to-MCP integration flows to translate REST/OpenAPI-based services into standardized MCP interfaces.
- Owned and resolved **15+ high-priority (P1/P2) tickets** across distributed components during a **live GA production war-room**, coordinating fixes across 4+ cross-functional engineering teams under **Scrum/Agile**.

***Siemens Healthineers***
**Technical Intern** **May - August 2025**
- NLP automation: Developed a **LangGraph workflow** to convert natural language user stories to **orthogonal array test cases**.
- MCP Integration: Automatically retrieved **Azure DevOps** metadata to integrate context with LangGraph agents.
- Impact: Reduced test suite size by 70% while preserving pairwise test coverage for complex diagnostic configurations.

***Centre for Data Science & Applied Machine Learning, PESU***
**Research Intern** **June - July 2024**
- Pattern Recognition: Built ML pipelines using XGBoost and Random Forest to classify and characterize fabric materials from cellular imaging data, improving feature extraction reliability for large-scale recycling workflows.

## Projects

- **End-to-End Rating Prediction and Recommendation System- CMU Project** **June - July 2026**
  Built a full ML pipeline (EDA → TF-IDF feature engineering → model selection/evaluation → deployment) to predict product star ratings from free-text reviews, deployed on Azure.
- **Deloitte Capstone Project - Industry Sponsored Project** **January - May 2025**
  Implemented a Retrieval-Augmented Generation (RAG) pipeline to verify credibility of employee-generated claims using evidence retrieval paired with explainable AI (XAI) outputs.
- **Kafka-Based Distributed Task Queue (YADTQ)** **November - December 2024**
  Built a fault tolerant, scalable, load balancing queue across multi-node workers using partitioning and topic management.

## Research Publications and Accepted Papers

- "ComicX: A Cognitive Pipeline for Reconstructing Comic Semantics into Emotionally Resonant Narratives"
  27th Conf. on Engineering **Applications of Neural Networks** (EAAAI/EANN), July 2026.
- "BrandGenesis: **Agentic Orchestration** of Multimodal Latent Spaces for Structured Brand DNA Induction via **RAG**"
  10th Int'l Conf. on Innovation in Artificial Intelligence (ICIAI), April 2026 (Accepted and Presented).
- "Profiling Synthetic and Non-Synthetic Fabrics Using a Cellular Device: **Machine Learning and Deep Learning Methods**"
  14th Int'l Conf. on Frontiers of Intelligent Computing: Theory and Applications (FICTA), July 2026 (Accepted and Presented).

## Activities and Awards

- Recipient of MRD, CNR **Scholarships** (50%, 40% tuition fee reimbursement respectively) based on academic merit.
- Top 10 in National level XR **Hackathon**, Waves Summit (India). XR project featured in *The Pioneer* newspaper and *DD News*.
- Former **Student Head, Lead Organiser** and Licensee, TEDxPESU; Event Organiser, ACM-W PES Chapter.